

Housing Discrimination and Algorithmic Bias

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Introduction and Historical Context

Racial covenants were legally binding clauses written into property deeds during the early 20th century that explicitly prohibited the sale or occupancy of homes by non-white residents. In Minnesota, racial housing covenants became widespread beginning around 1910, when developers and real estate agents in Minneapolis and surrounding areas began inserting deed restrictions that explicitly prohibited property sales to non-white residents. According to the Minnesota Historical Society, these covenants were used extensively across both Hennepin County and Ramsey County, shaping patterns of residential segregation that continued to influence neighborhood demographics and property values long after being outlawed. Thousands of these covenants shaped patterns of segregation that affected neighborhood investment, property values, and generational wealth. Although these practices were outlawed under the Fair Housing Act of 1968, the effects persist in the economic and spatial patterns visible today. As modern mortgage lending increasingly relies on algorithmic systems to assess credit risk, important ethical concerns come to mind. Investigations such as *Machine Bias* by ProPublica demonstrated how seemingly neutral risk algorithms in the criminal justice system disproportionately harmed Black individuals, revealing how biased historical data can translate into discriminatory outcomes. Similarly, in her TED Talk, Joy Buolamwini exposed racial bias embedded in facial recognition systems, showing that algorithms can reproduce structural inequalities even without explicitly using race as a variable. These cases raise a critical ethical question for housing finance; if modern models rely on data shaped by decades of legally enforced segregation, can they actually operate fairly? This project examines whether historical housing discrimination in Hennepin and Ramsey Counties continues to influence present day lending patterns through data bias and algorithmic bias, connecting past injustice to contemporary ethical challenges in predictive technology that we are using today.

Methodology

Data Collection

For the Hennepin data, we used the data table provided but noticed it needed to be cleaned to be ready for analysis. To start analyzing and modeling we needed to clean the Hennepin county data using the provided jupyter notebook for housing discrimination. The provided Jupyter notebook allowed us to retrieve results similar to the data that was given to us but very little cleaning was completed so we started there. Our primary objective in this step was to help fill missing values that were initially present in the data, specifically the *census_tract*, *property_value*, *income*, *interest_rate*, and *covenant_density* features. We converted the census tracts to eleven-character strings rounded off to match the census tracts in the initial database. We then dropped any null census tract values because we did not want to fill these with superfluous census tract values. To deal with the property values and income, we decided to fill the missing values with the median value for both columns. Lastly, we filled the covenant density missing values with 0 because these are areas without any covenants. We wanted to clean the

Ramsey data that we acquired in this same process, but some additional steps were required to get it ready for modeling and analysis. The data from the mappingpredjudice.umn.edu website and unlike the Hennepin county data that was processed the Ramsey County data was still in progress in uploading and processing the table. So the data acquired was not formatted the same as Hennepin county and did not have X and Y coordinates for each of the housing covenants. So to fix the coordinate issue the group used ArcGIS pro software and downloaded the shapefile for Ramsey county from github. The shapefile contains 5 files that contain the geometry, index file, attribute table, and a coordinate projection and was uploaded on ArcGIS to show the county on a map. The shapefile needed to be updated with x and y coordinates by calculating the geometry for each address's centroid. Claude AI was used to match the same table as Hennepin county with making sure the projected coordinate system was correct and all column names were the same. Then used the provided code, Housing Discrimination and Algorithmic Bias: A Data Ethics Project, from how to clean the Hennepin county data and used it for the Ramsey county cleaning. Now the Ramsey county data was ready to use for modeling and analysis. Before we could use the Ramsey data we needed to clear up some missing values in the same way we did for the Hennepin data. The data was fairly similar and demonstrated similar imbalances which were later reflected in the bias analysis of both datasets.

Analysis

For our analysis on both the Hennepin and Ramsey county mortgage loan datasets, we tailored our data exploration, scaling, modeling, and visualizing to the project objectives and overview. To give a brief overview, we started by examining and analyzing the relationships between covenant density and property values. We then dove deeper on mortgage loan approval patterns and how they relate to location and race. After this step, we discovered some surprising proxy variables which are associated with historical discrimination. Lastly, we covered and compared lending patterns and how they differ in covenanted and non-covenanted areas. To dig into the data bias that exists within both data sets we looked at the average property values and incomes across the covenant density groupings as well as looking at the representation of each race within the dataset and each covenant density group. For the algorithmic bias analysis, we created multiple hands-on logistic regression models which utilize a system similar to most lending agencies. First, we created a baseline logistic regression which didn't utilize any of the covenant count or covenant density to exemplify the racial features. Then, we created two logistic regression models which incorporated the covenant data in different ways. Lastly, we created a logistic regression model that incorporated an interaction effect between african american as a race and being located in an area that has high and very high covenants.

Findings

Data Bias Analysis

In the Hennepin data analysis there were interesting findings that do show some data bias in the data itself. In the data there is a racial skew for loans for both counties with white is 63% of the

demographic for each county with asian and black at 5.9%, native american at 0.7% and pacific islander and multiracial at 0.1% of the data. The biggest issue is the unknown race at a high demographic of 22.6% and this is data bias from not knowing the race could help or make the bias worse if the information was available.

This is also shown from a historical context: the data was already skewed from racial bias using the covenants from the early 1900s when the covenants started to take place. The data shows there is some influence of the covenants still having effect still even after the 1968 housing act made it illegal.

There is also income bias with an unknown of what race having a lot of outliers on the boxplot of Income by Covenant Density (Log Scale) where there may be more or less bias towards rich people by not knowing the race of them skews the data by not knowing what race they are. This unknown of race can disproportionately affect the covenant density towards a certain race with saying there is no racial bias towards anyone even in the highest density covenants with regards to income levels of the covenant. The same goes for low income as well showing they might be getting loans in the covenants compared to the wealthy. This unknown of the race is also skewing the analysis by not showing the true race equality or racial bias in each of the covenants in regards to loan approvals. The same goes for Ramsey county and it follows the same skewness of race and the unknown race as well.

Algorithmic Bias Analysis

The demographic and the loan acceptance rate go hand in hand in showing there is racial bias in giving out loans. The Hennepin county heat map shows the racial bias for mortgage approvals where white having a average of ~77% approval rate across all covenant densities compared to black people having the lowest approval rate with an average of around ~59% approved. Only joint, asian, and multiracial having slighter approval rates in the medium and high covenant densities. The Ramsey county heatmap for the mortgage shows native americans high high loan approval in the medium to high covenant density with even higher at times compared to white at ~77%. The biggest is racial approval rate is between white, asian, and black where black has the lowest approval rating for all covenants with asian slightly better approval rates. In comparing the counties there is a slightly better approval rate in Ramsey county rather than Hennepin county.

The other heatmap of Approval Rate Advantage/Disadvantage really shows the racial bias in approvals. where native american, black, and pacific islander having the greatest disadvantage in Hennepin county while in Ramsey county black people are the biggest disadvantage across the board. The interesting and highest disadvantaged are Multiracial people in medium density in Ramsey County at -61% which is the lowest for both counties.

County Comparison

The results from Hennepin and Ramsey looked very similar overall. In both counties, White applicants had the highest approval rates, while Black applicants were the lowest. Even after adjusting for things like income, loan amount, and tract level factors, race still showed up as important in the models. Also, neighborhood factors like tract income and minority population were clearly tied to approval outcomes in both places. Seeing the same pattern in two different

counties makes it less likely that this is random. It points more toward a bigger issue. Even though Hennepin and Ramsey are different areas, they both seem to reflect the same housing history, and that history is still showing up in how lending works today.

Discussion

Implications for Modern Lending

The findings from both Hennepin and Ramsey Counties suggest that modern lending systems are not as neutral as they may appear. Even though today's models do not explicitly include race as a variable, they rely heavily on factors like income, property value, and geographic location, all of which are deeply tied to historical patterns of segregation created by racial covenants. Because of this, the models can end up reinforcing the same inequalities they are supposed to avoid. One of the biggest implications is that algorithmic decision-making in lending can unintentionally reproduce historical discrimination at scale. A model trained on biased historical data will learn those patterns and treat them as "normal," which means certain groups may continue to face lower approval rates even when controlling for financial factors. This becomes especially concerning when these systems are used widely across institutions, because small biases can turn into large, systemic disparities.

Another important implication is that relying on model accuracy alone is not enough. A model can perform well statistically while still producing unfair outcomes across different demographic groups. This challenges the idea that objectivity in data automatically leads to fairness. Instead, lenders need to actively evaluate both performance and equity at the same time. These findings also highlight the need for stronger regulatory oversight and ethical standards in financial modeling. If lenders continue to use variables that are closely tied to historically segregated neighborhoods, they risk embedding past discrimination into future decisions. Without intentional intervention, technology may not fix inequality, it may simply modernize it.

Proposed Mitigation Strategies

One thing lenders need to actually commit to is running regular fairness checks. That means tracking approval rates, denial rates, and interest rate spreads broken down by race and neighborhood over time. If a model has high accuracy but still denies certain groups more often, then something is off. Accuracy alone should not be the goal. Fairness should be looked at alongside it. If improving one makes the other worse, that is something that needs attention, not something to ignore.

Another area worth looking at more carefully is which variables are even being used in the first place. Features like neighborhood income, property value, and location can look totally normal, but in this data they are directly tied to historical discrimination. So before using those variables, questions should be asked. Lenders should ask if they are really measuring creditworthiness or just reflecting old patterns in a different way.

Transparency is something that does not get enough attention. If a loan officer or even the compliance team cannot explain why someone was denied, that is a real issue. It is not just

about having a working model. People should be able to understand how decisions are made. If the process is unclear or hard to follow, then it becomes difficult to question decisions or fix what is going wrong. Being able to explain the reasoning is what makes accountability possible.

Limitations and Future Work

There are several limitations in this analysis that should be considered when interpreting the results. One of the biggest challenges was the high percentage of “unknown” race values in both datasets. With over 20% of applicants missing race information, it becomes difficult to fully understand the extent of racial disparities. This missing data could either hide or exaggerate bias, depending on how those unknown values are actually distributed. Another limitation comes from the differences in data quality between Hennepin and Ramsey Counties. While the Hennepin dataset was more structured, the Ramsey data required additional processing, including geospatial adjustments using ArcGIS and matching formats across datasets. These extra steps introduce potential sources of error, especially when calculating coordinates or aligning features across different data sources. Additionally, the analysis is limited to the variables available in the dataset. Important factors such as credit score, employment history, or debt-to-income ratio were not included, which means the models cannot fully capture all aspects of lending decisions. While this does not eliminate the observed disparities, it does mean that there may be other contributing factors not accounted for in this analysis.

For future work, one key improvement would be incorporating more complete and detailed data, especially reducing the number of unknown race entries. Expanding the analysis to include additional counties or states would also help determine whether these patterns are consistent across different regions. Another important direction would be testing alternative modeling approaches that explicitly account for fairness, such as fairness-constrained models or bias mitigation techniques. Comparing these models to traditional approaches could provide insight into how lending systems can be improved without sacrificing performance. Finally, future research could explore the long-term impact of these disparities by connecting lending outcomes to wealth accumulation, homeownership rates, and neighborhood development. This would help further illustrate how historical discrimination continues to shape economic opportunities today.